https://github.com/Et1enneee/comp3500-logistics

COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. The local application serves one warehouse using Next.js, TypeScript, Prisma and SQLite. Real GPS, barcode scanning, recipient telephone collection, payment and carrier integration are outside scope. Deployment is local-only; data integrity is enforced by transactional guards (§5). Requirements and models link to evidence; §10 collects the trail.

2. Development process

Plan-driven increments have triggers and verified outcomes; integration and continuous integration provide feedback. Increments cover product and specification changes and recorded process milestones.

Table A. Increment, trigger and outcome

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Project selection	Responsibilities and contracts recorded
Test-led review, 17 Oct 2026	Assess order and stock functions	Defects recorded; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Failed endpoint and missing ledger	Transaction verified
Integrated application, 1 Nov 2026	Integrate foundation	Customer, staff and admin workflows
Driver and privacy, 2 Nov 2026	Delivery and data minimisation	Assigned tasks and privacy checks
Documentation hardening, 10–11 Nov 2026	Model and evidence review	Diagrams and traceability aligned
Export simplification, 11 Nov 2026	Scripted walkthrough finding (issue #15)	Selected-date export in three actions; regression passed

The workflow, issue tracker and increment register record integration, problems and context.

3. Requirements

The specification defines obligations and rationale.

Functional requirements and verification

FR	The system shall	Rationale	Verification
FR-01	create orders with product lines and totals	start fulfilment	Test
FR-02	enforce lifecycle and stock reservation or release	protect invariants	Test
FR-03	number orders and shipments and show owner tracking	retrieve progress	Test
FR-04	show inventory, replenishment, ledger (inventory movement log) and low-stock alerts	support replenishment	Test
FR-05	record event notifications and allow reading	communicate progress	Test
FR-06	provide admin aggregates and CSV export	support review	Test
FR-07	authenticate and enforce role and object access	protect data	Test
FR-08	assign shipped orders and simulate delivery	complete handoff	Test

NFR-01 performance (API p95 ≤2 s, first paint ≤3 s); NFR-02 integration availability (40 of 52 recorded runs green); NFR-03 least privilege (role gates on every mutating endpoint); NFR-04 ≤3 actions per scripted operation; NFR-05 process quality (CI-verified increments).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Treatment status
RK-01 scheduling slip	Increment register	Managed — scheduling plan
RK-02 contract rework	FR-02, NFR-05	Mitigated — transition tests
RK-03 test-week capacity	Priority register	Managed — scope priorities
RK-04 setup delay	NFR-05	Managed — CI workflow
RK-05 demo data	FR-01/04/06, NFR-05	Managed — seed regression test
RK-06 assignments and test squeeze	Risk plan	Managed — assessment schedule
RK-07 stack unfamiliarity	NFR-05 support	Managed — setup guide
RK-08 report and video compete for time	Risk plan	Managed — report and video plan

4. System models

Five views link roles, lifecycle, data, transaction, layers. Local transactions preserve consistency without splitting services; state and stock guards enforce rules.

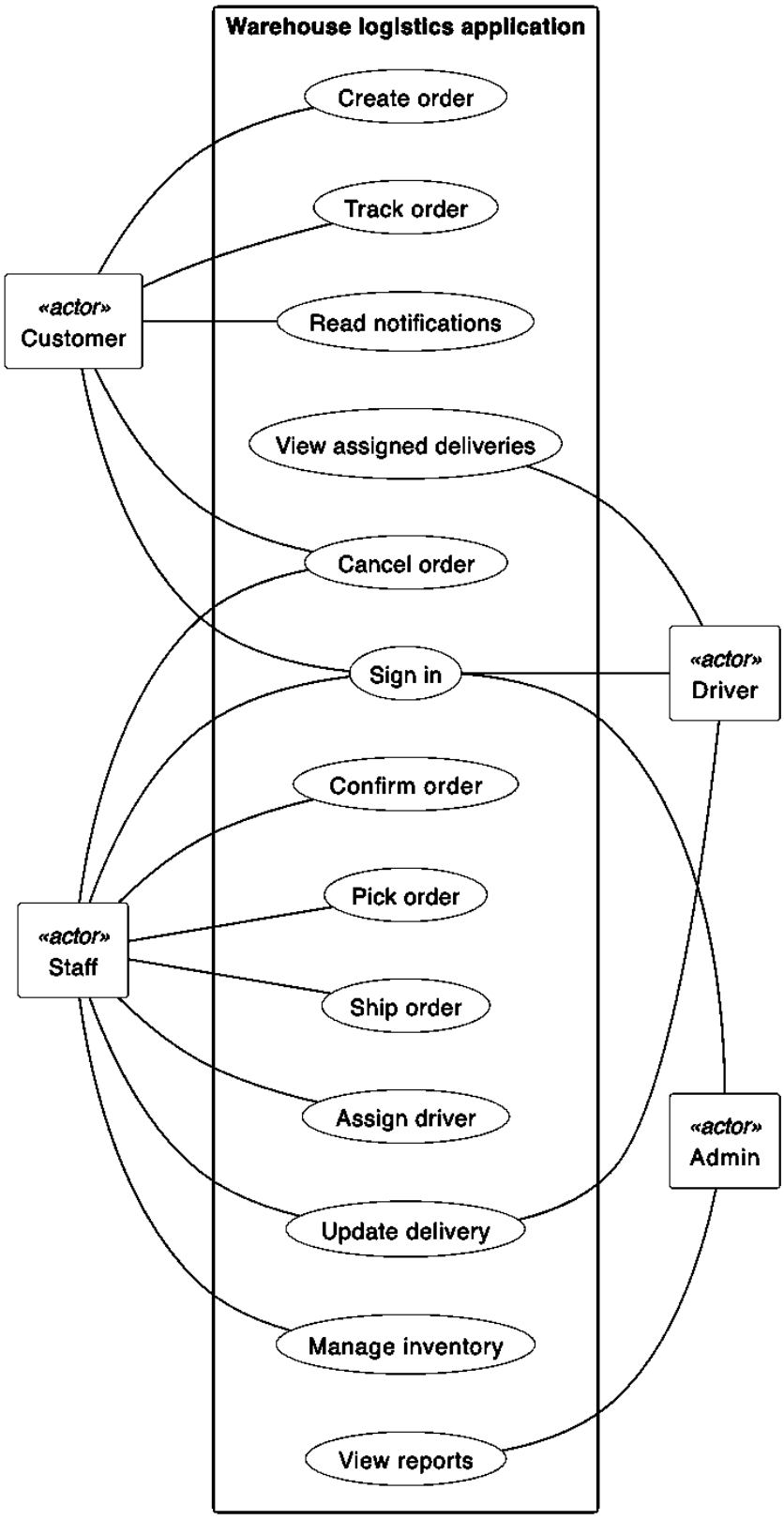


Figure 1. Which roles perform each function? Source.

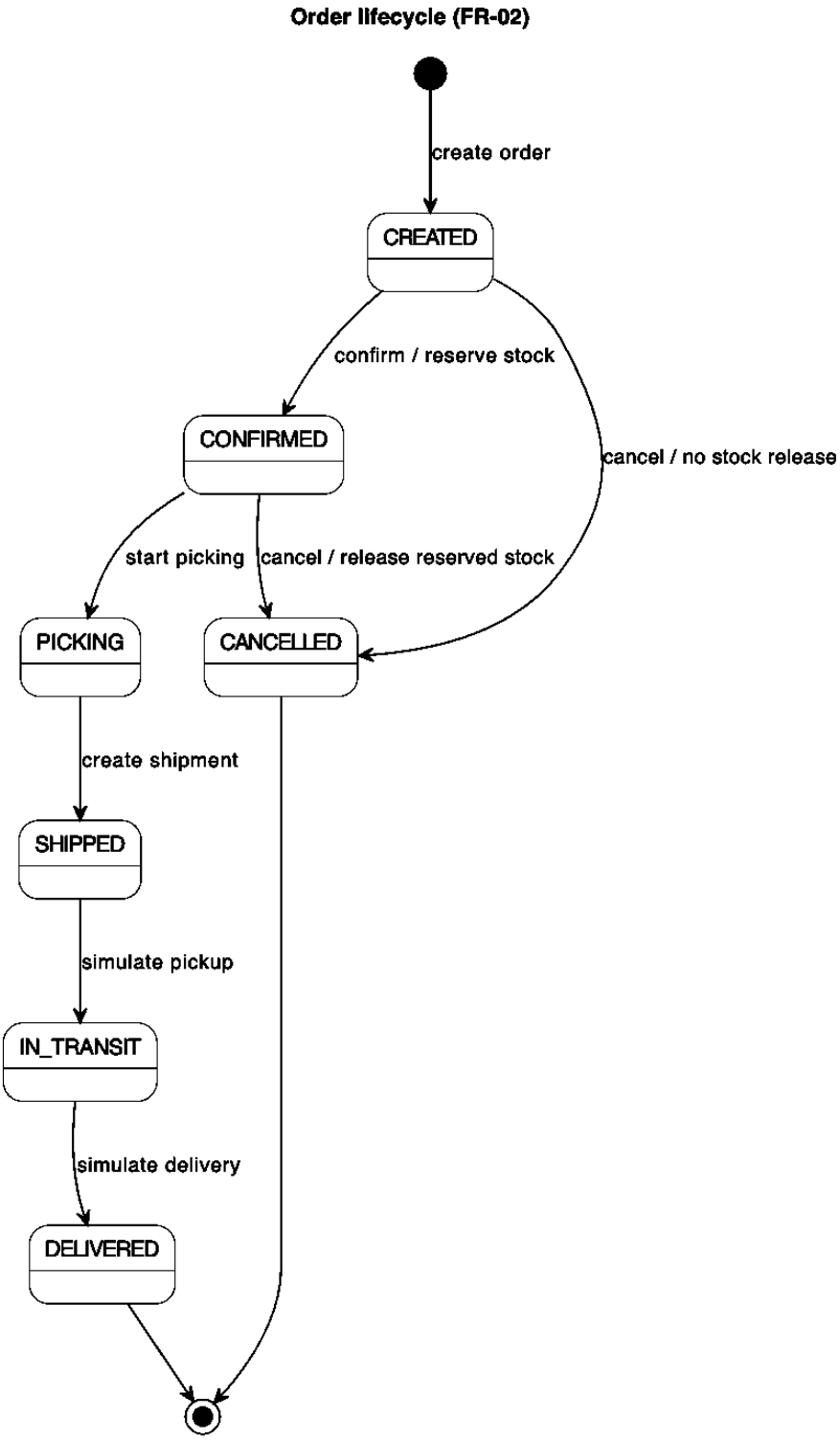


Figure 2. Which order transitions and stock effects are valid? Source.

Each item references one product; each order has 0..1 shipment; each product has 0..* inventory logs.

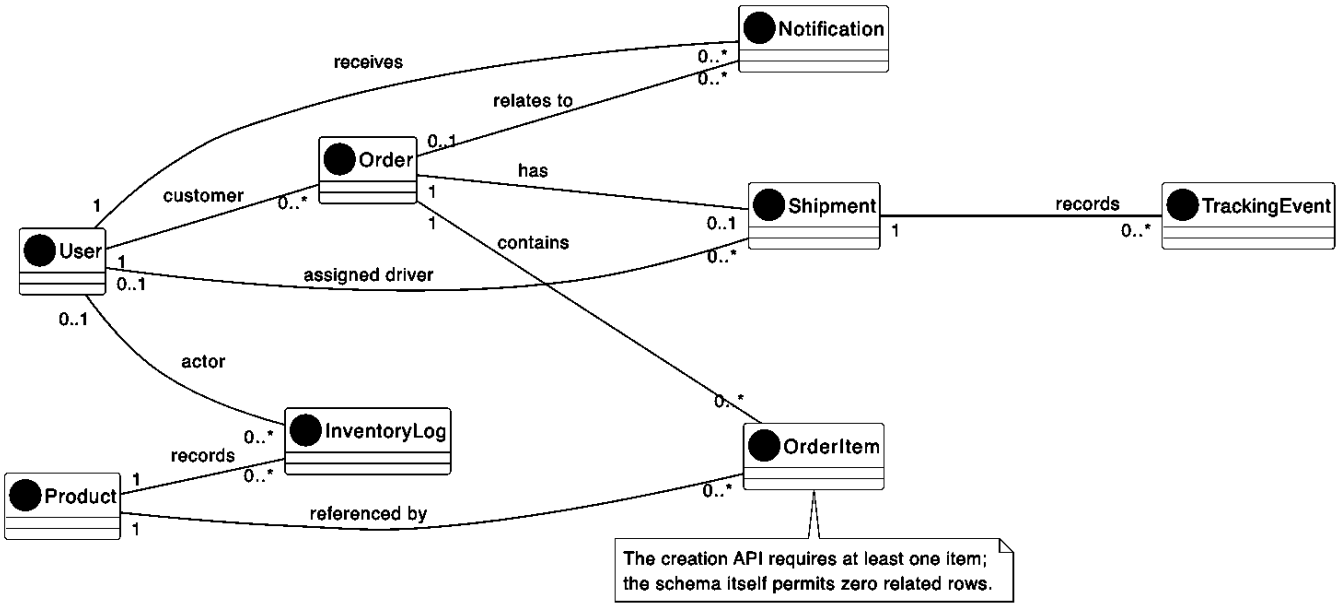


Figure 3. Data relationships: each order requires at least one item, enforced at the creation API. Source.

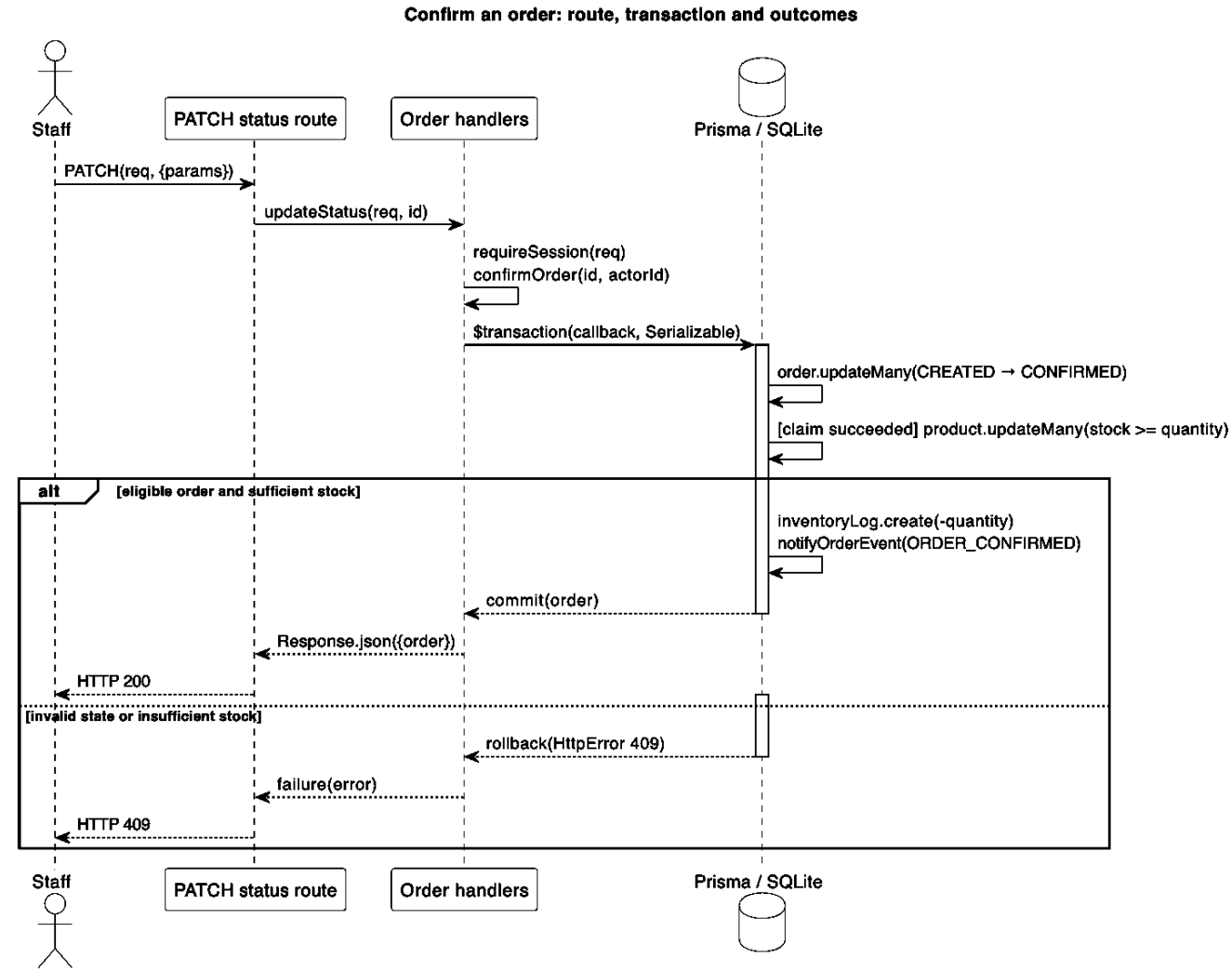


Figure 4. How does confirmation commit or roll back? Source.

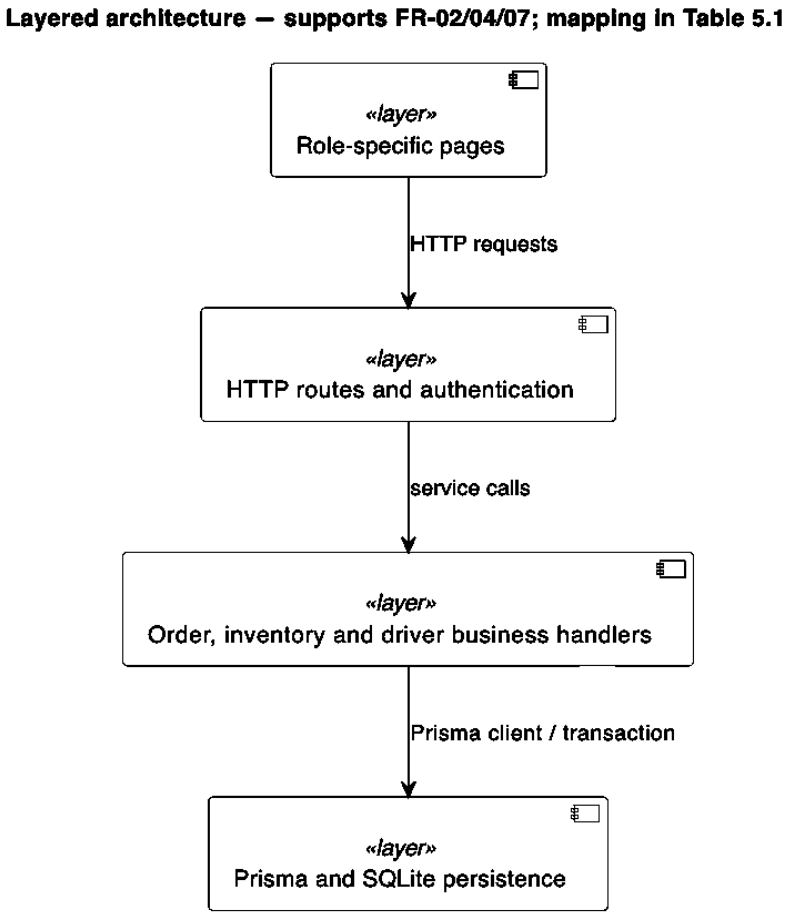


Figure 5. How do four layers connect? Source.

5. Implementation

Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED, then IN_TRANSIT and DELIVERED; CANCELLED per the transition table. Customers cancel their own CREATED or CONFIRMED orders; staff cancel CONFIRMED orders; stock is restored only on CONFIRMED cancellation.

Design	Implementation	Verification
Guards: CREATED and sufficient stock	order-api.ts:250–253, 276–280	Orders
Serializable state, stock, ledger and notification writes	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; retained after delivery	driver-api.ts:44–51; transition	Driver
Selected-date export	admin/page.tsx:24–48	Browser

6. Verification and validation

Verification checks requirements; validation assesses fitness for users. The regression log and CI show 38 unit, 140 integration, 2 delivery (driver simulation) and 26 browser checks; suite breakdown in the regression log.

Boundary case	Observed	Test
Repeat confirmation	HTTP 409; state and stock unchanged	Orders
Stock equals demand	Confirmation succeeds; stock becomes zero	Orders
Stock 4 versus threshold 5	Alert; stock 5 is not low	Inventory

Insufficient stock returns HTTP 409 with unchanged state and stock. In a local-loopback measurement recorded on 11 November 2026, API p95 was 3.2–13.6 ms and first-contentful-paint p95 was 64–96 ms, against NFR-01 targets of at most 2 s and 3 s respectively. NFR-02 records 40 successful build–test–browser runs (observations).

7. Evolution

The early defect record drove the confirmation transaction and retest; the later audit corrected models and renewed tests. A four-action administrator task became three actions including date entry (regression).

8. Validation

The automated walkthrough validates eleven core operations across all four roles, at most three scripted actions each, with acceptance criteria fixed before the run; results are logged with the run.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Evidence
Diagrams	Corrected models and architecture	Compliance matrix
FR-04	Stock transition effects belong to FR-02	Traceability
Increments	Triggers and outcomes	Register
Early performance figures	Superseded by the recorded 11 Nov measurement	Measurement
Admin task	Export re-tested; scripted walkthrough records the task (§8)	Issue #15
Unit count	Two availability-summary regression tests increased unit count from 36 to 38	Test history
NFR scope	Criteria revised for local deployment	SRS revision log

Both goals connect order decisions and role-separated workflows to evidence.

10. Evidence

Traceability · Specification · Evidence directory · Baseline record.

Full evidence trail: docs/final-report.md in the repository (evidence baseline `6eb01afb1798c7361205b10597d80c0c26e8fdf6`; code, evidence and test logs at this commit); system version v0.4.2.